Intelligent SFP module ISM4120I
User Guide

Version 3, 2025

Software version 1.4.5

Table of Contents

1. Safety measures 5
2. Introduction 6
   2.1. Overview 6
   2.2. Main features 6
      2.2.1. Debian 12 operating system 6
      2.2.2. DPDK framework support 6
      2.2.3. Remote management 6
3. Technical specs 7
   3.1. General 7
   3.2. Operating conditions 7
   3.3. External module view 8
   3.4. Internal block diagram 8
   3.5. DIP switch 9
   3.6. LED 9
4. Delivery set 10
5. Software structure 10
6. Installation and setup 11
   6.1. Inserting Smart SFP into a Host Device 11
   6.2. IP interfaces 12
   6.3. Default accounts 12
   6.4. Connect to Smart SFP 12
      6.4.1. SSH 12
      6.4.2. SSH connection using a key 13
7. Connection schemes of the module 15
   7.1. Out-of-line connection 15
      7.1.1. Line interface is not connected 15
      7.1.2. Line interface is connected 16
   7.2. In-line connection 17
      7.2.1. Using bridge for data transfer 17
      7.2.2. Using routing for data transfer 17
8. Examples of typical settings 19
   8.1. Disabling RX_LOS signal generation 19
   8.2. Setting up DNS server address 19
   8.3. Installing Docker 20
9. Factory reset 21

10. Software upgrade 22
10.1. Upgrading from software versions 1.4.5 and later to newer ones 22
10.1.1. Necessary files 22
10.1.2. Keeping settings during software upgrade 22
10.1.3. Upgrade procedure 23
10.2. Upgrading from software version 1.4.2 to 1.4.5 24
10.2.1. Necessary files 24
10.2.2. Module settings after upgrading from software version 1.4.2 to 1.4.5 24
10.2.3. Upgrade procedure 24
11. EEPROM memory reprogramming 26
11.1. EEPROM reprogramming using a programmer 26
11.2. EEPROM reprogramming without using a programmer 27
12. Maintenance 28
Appendix 1. Pins of the electrical connector of SFP module 28

1. Safety measures

Before using the device, carefully read this user manual.

- Get the device from the box and make the external inspection.
- Keep the device in normal environmental conditions for at least 2 hours (if the device has been previously kept in conditions distinct from normal).
- When using the device, it is necessary to comply with the general requirements of fire safety regulations.
- It is necessary to protect the device from shock, moisture, and dust, prolonged exposure to direct sunlight.

2. Introduction

2.1. Overview

Intelligent SFP module (hereinafter may be referred to as device, module, Smart SFP) is a network appliance in the SFP form factor. Smart SFP can be installed in any network device with an SFP slot. Smart SFP is running Debian Linux, so that you can run on it big variety of applications for different purposes.

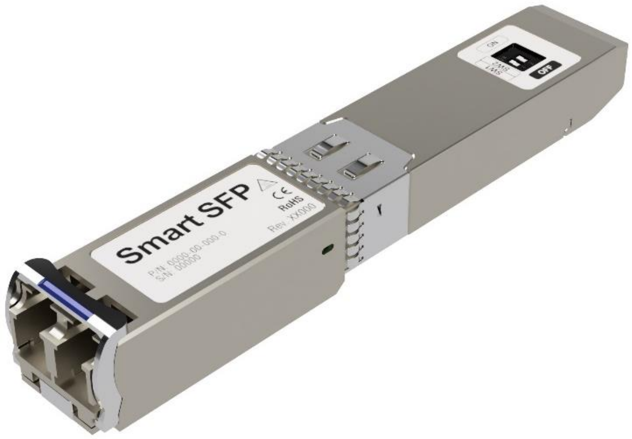

Figure 2.1. Smart SFP appearance

2.2. Main features

The current software version of Smart SFP includes the following features:

- Debian 12 operating system
- DPDK framework support
- Remote management

2.2.1. Debian 12 operating system

The module has Debian 12 operating system with Linux Kernel 6.1.x. It allows to install any compatible utilities.

2.2.2. DPDK framework support

The module has a built-in DPDK 24.11 framework support.

2.2.3. Remote management

The module supports remote management:

- CLI via SSH
- VLAN-based

3. Technical specs

3.1. General

Table 3.1. Common specifications

Parameter	Description
Manual Switches	2 x DIP Switches for User Settings
CPU	Marvell Armada 3720 ARMv8 dual-core, 1,2 GHz
RAM	1 GB
Nonvolatile storage memory	4GB eMMC
Availiable storage on / partition	569 MB
Availiable storage on /home partition	99 MB
OS	Debian bookworm 12, Linux kernel 6.1.x
Indication	1×LED
Weight	33 g
Power consumption	up to 3.5 W, depending on use case
Interface	1000BASE-LX
Data Rate	1.25 Gb/s
Wavelength	1310 nm
Cable Type	SMF
Max Distance	20 km
Tx/Rx type	FP/PIN
Transmitter Power	-4.5 ~ -3.5 dBm
Receiver Sensitivity	< -19 dBm
Receive Overload	-3 dBm
Connector	LC
Digital Diagnostic Monitoring	Yes
Dimensions	W: 13.9 mm / H: 12.3 mm / L: 79.7 mm

3.2. Operating conditions

Table 3.2. Operating conditions

Parameter	Description
Operating temperature	-25...+85 °C
Storage temperature	-40...+85 °C
Relative humidity	20-85%, non-condensing

Taking into account the fact that the module is not a self-sufficient device and is operated as part of other equipment, the specified operating temperature range of the module does not indicate the range of ambient temperatures at which the module operates correctly. This range indicates the temperature values at which the module components operate correctly.

The ambient temperature range is not specified, because the module temperature may differ significantly from the ambient temperature because the module temperature is affected by many factors, such as the type of cooling in the room, the type of equipment where the module is installed, the type and design of cooling in the equipment, the location of the equipment in the rack, the presence or absence of neighboring SFP modules, the current temperature of neighboring SFP modules, etc.

3.3. External module view

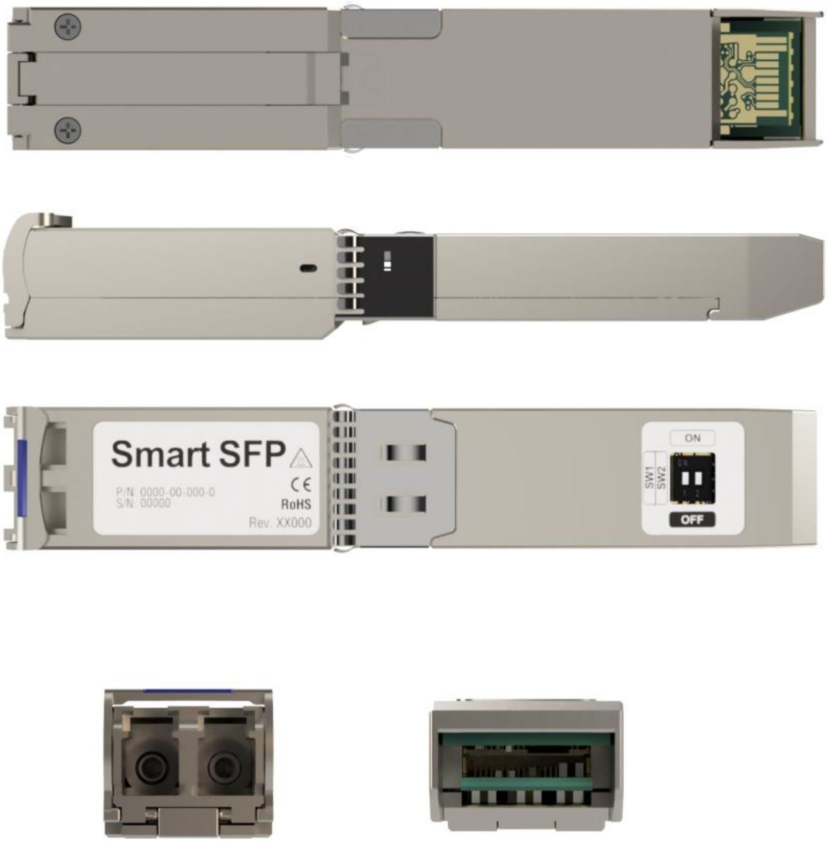

Figure 3.1. Smart SFP appearance

3.4. Internal block diagram

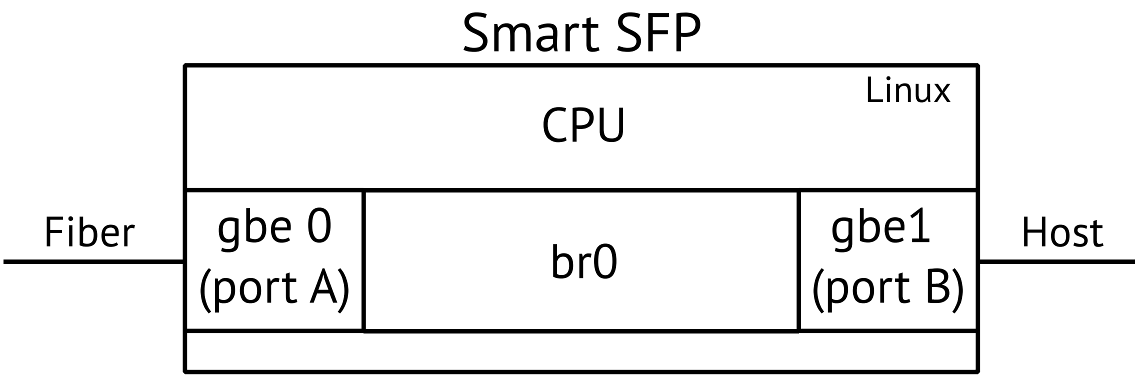

Figure 3.2. The module internal block diagram

3.5. DIP switch

Smart SFP includes a two-part DIP switch on its upper part, used to select the device's operating mode.

Note. DIP switch control is not supported in the first software version.

Figure 3.3. Smart SFP DIP switch

Table 3.3. DIP switch positions and operating modes

	SW1	**SW2**
Regular mode	ON	ON
Recovery mode	ON	OFF
Not used	OFF	ON
Not used	OFF	OFF

3.6. LED

Figure 3.4. Smart SFP State LED

State LED indicator is located on the side of the module. LED color depends on the operating mode and the device state:

1. Regular mode:
   - After the module is inserted into the host device, the LED indicator blinks during software initialization for about ten seconds
   - If an error occurs, the indicator lights up red.
2. Recovery mode:
   - Starting service firmware: does not light for about 10 seconds, then blinks green
   - Writing software image to ROM: flashes red for about 15 minutes
   - Constant color – writing is completed, you can eject the module from the host equipment. "Green" indicates successful completion of the writing process, "red" indicates an error during the writing process.

4. Delivery set

Smart SFP delivery set includes:

- The module
- Quick start guide
- Packing box

Packing box size – 120×75×25 mm. Packing box weight with the module – 58 g.

5. Software structure

The device is running under Debian OS, Linux kernel 6.1.x. Standard OS commands and configuration files are used to configure network and system parameters. For example, network interfaces configuration is performed using the “ifconfig” command. If you want to save the settings after a reboot, change the file /etc/network/interfaces.d/gbe. Detailed information about commands usage can be obtained through the built-in help system or at www.debian.org.

Debian OS's allows the installation of standard Linux packages on the Smart SFP. For that:

1. Remount the Smart SFP file system with write permissions:

   mount -o rw,remount /

2. Install standard software.
3. After the installation of the software is completed, the file system should be remounted with read-only rights:

   mount -o ro,remount /

6. Installation and setup

6.1. Inserting Smart SFP into a Host Device

Smart SFP generates a lot of heat while operating. Because of that it is important to follow these recommendations:

1. Use Smart SFP only in host devices with active cooling system.
2. Do not install Smart SFPs in adjacent SFP slots
3. Do not install Smart SFP in SFP slot adjacent to high powered long distance SFP modules.

You need to insert Smart SFP into the host device to configure it via the host or operate it. Smart SFP is a hot-swappable device – you do not have to switch off the host unit when inserting or extracting it:

1. Insert Smart SFP into an MSA-compatible SFP/SFP+ socket of the host equipment. Press Smart SFP firmly into the SFP socket until Smart SFP clicks into place.
2. Connect cable.
3. Smart SFP is ready to operate.

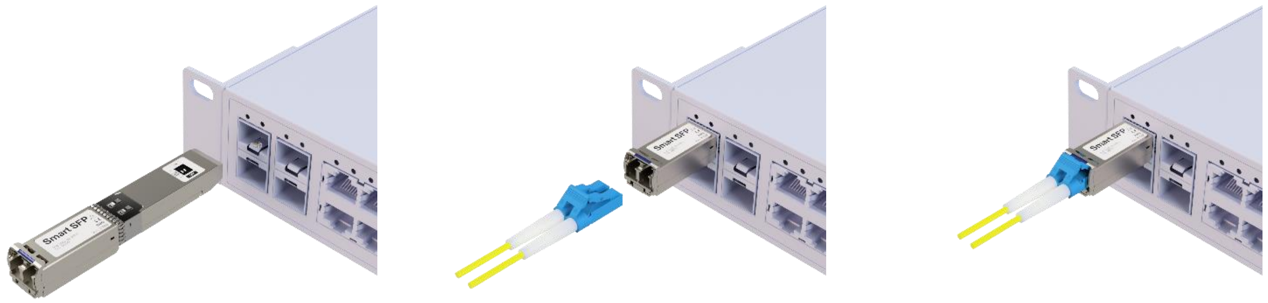

Figure 6.1. Inserting Smart SFP

To eject Smart SFP (see Figure 6.2):

1. Disconnect any cables attached to Smart SFP.
2. Push the release latch on Smart SFP's underside in the direction of the host device.
3. While continuing to keep pressure on the latch, pull Smart SFP gently out of the socket. Release the latch after you have completely removed Smart SFP from the socket.

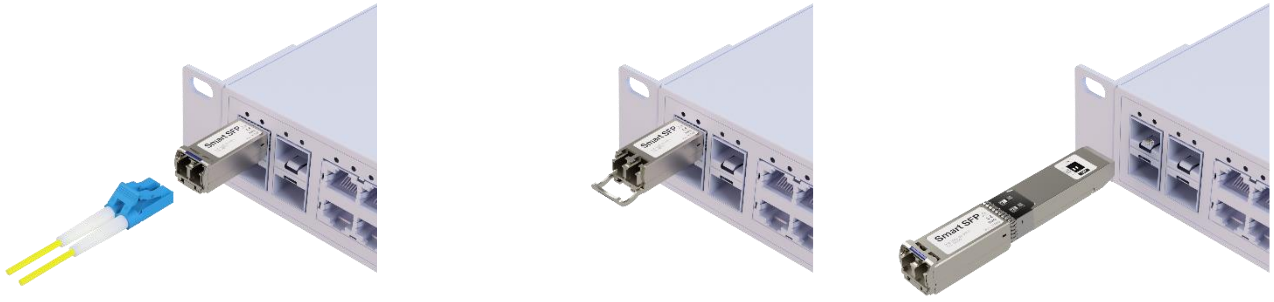

Figure 6.2. Ejecting Smart SFP

6.2. IP interfaces

- Users can get access to the device using the SSH protocol.
- Interfaces A (SFP) and B (HOST) are detected in the device's Linux system as gbe0 and gbe1 correspondingly.
- Interfaces A and B are combined into the network bridge interface. In the operating system, the bridge is detected as "br0" and has a default IPv4 address: 192.168.1.1.
- By default, the module is configured with a default gateway: 192.168.1.254.

6.3. Default accounts

There are two different types of user accounts on the device: root, user.

Table 6.1. Default accounts

Login	**root**	**user**
Password default	PleaseChangeTheRootPassword	PleaseChangeTheUserPassword
Description	Standard Linux account with all rights and permissions	Standard Linux account
Shell default	/bin/bash	/bin/bash
SSH	Key-based authentication only	Key-based and password authentication

6.4. Connect to Smart SFP

6.4.1. SSH

To connect to the module via SSH, you must first configure the IP address of the personal computer (PC) so that the PC and the module are on the same subnet. You can connect via port A and port B.

6.4.1.1. Windows

1. Connect the module.
2. Open a terminal program with SSH support (for example, PuTTY or HyperTerminal) on a PC.
3. Set the IP address of the module and open the connection:

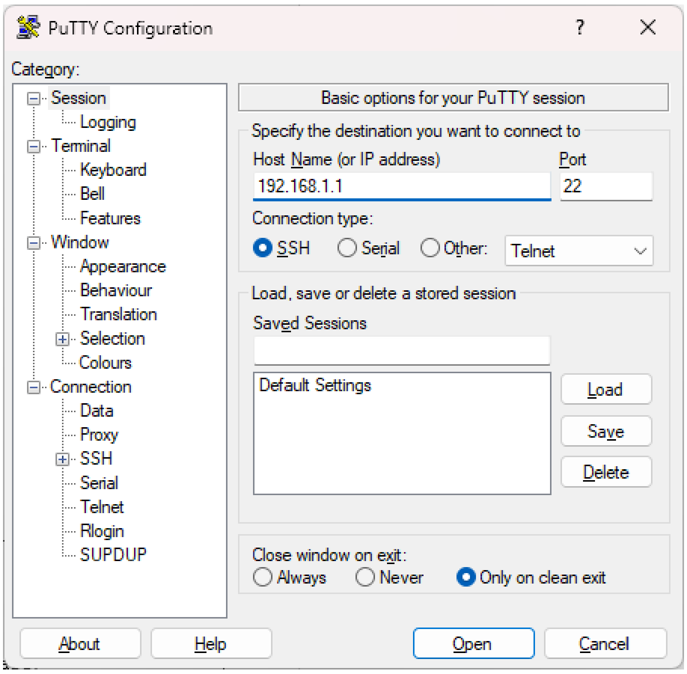

Figure 6.3. SSH connection

4. Enter a username (“user”) and password (see Section 6.3).

```
login as: user
user@192.168.1.1's password:
Linux  smart-sfp  6.1.67mtk-0.0.12-g077ec6d58fc5  #1  SMP  PREEMPT  Mon  Jun   2  15:22:04 MSK 2025 aarch64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Thu Mar  6 15:00:01 2025 from 192.168.1.111
user@smart-sfp:~$
```

Figure 6.4. Successful connection

6.4.1.2. OS Linux

1. Connect the module.
2. Connect from PC via SSH, specifying the account (user) and the IP address of the module. For example:

   ssh user@192.168.1.1

3. If the connection is successful, the program will prompt you to enter a username and password (see Section 6.3).

6.4.2. SSH connection using a key

6.4.2.1. Installing a key on the module

To connect via SSH using a key, you need a pair: a private and a public key. You can use a pair you already have or generate a new pair on the computer from which connections using the key will be made. To install a public key on the module, you need to:

1. Connect to the module via SSH using the password.
2. Create a folder to store the authorization file:

```
user@smart-sfp:~$ mkdir .ssh/
```

3. Create a file responsible for authorization and enter the key text into it. The text in quotes should be replaced with the text of your public key in a similar format:

```
user@smart-sfp:~$ echo "ssh-ed25519
AAAAC3NzaC1lZDI1NTE5AAAAIEtsAiu+XtVxIhMR4/WT0jsvvbdx32/sT/fy+u03sj7m user@pc" >>
.ssh/authorized_keys
```

6.4.2.2. Connection using a key

To connect using a key, you need to pass the private key to your SSH client:

- when using PuTTY, you need to specify the path to the private key when setting up a connection in the menu Connection -> SSH -> Auth -> Credentials -> Private key;
- when using a standard SSH client in Linux, the path to the private key should be specified using the -i switch:

```
ssh -i /home/user/.ssh/my_private_key user@192.168.1.1
Linux  smart-sfp  6.1.67mtk-0.0.12-g077ec6d58fc5  #1  SMP  PREEMPT  Mon  Jun   2  15:22:04 MSK 2025 aarch64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Thu Mar  6 15:00:01 2025 from 192.168.1.111
user@smart-sfp:~$
```

7. Connection schemes of the module

The probe can be connected in two ways:

1. "Out-of-line" – there is no data transit over the module with default configuration.
    - The line interface can be disconnected. In such case access is only possible through port B. An IP address is assigned to interface gbe1 (port B).
    - The line interface can be connected. In this case access to Smart SFP is possible using the IP address of interface gbe1 (port B) from the Host side, and using the IP address of interface gbe0 (port A) from the Line side. The module will act like Linux device with two network connections.
2. "In-line" – the module can be configured to allow transit traffic:
    - Traffic transit can be set to work in L2 mode using the bridge. An IP address for access to Smart SFP should be assigned to the bridge interface.
    - Traffic transit can be set to work in L3 mode by enabling L3 forwarding. In this case access to Smart SFP is possible using the IP address of interface gbe1 (port B) and the IP address of interface gbe0 (port A).

7.1. Out-of-line connection

7.1.1. Line interface is not connected

Access to the module is only possible through port B in this connection and configuration option. No user traffic is transmitted through the probe.

Before configuring the module to operate in this mode, it is necessary to disable the generation of the RX_LOS signal (see Section 8.1).

To configure this operating mode, you should assign an IP address on the gbe1 interface (port B) and delete or comment out the lines with the br0 interface settings:

1. Connect to the device via SSH using the IP address of the br0 interface.
2. Log in as user and then switch to the root account:

```
user@smart-sfp:~$ su
```

3. Remount the file system for writing:

```
root@smart-sfp:/home/user# mount -o remount,rw /
```

4. Edit the /etc/network/interfaces.d/gbe file as follows:

```
root@smart-sfp:/home/user # vi /etc/network/interfaces.d/gbe
auto gbe0
allow-hotplug gbe0
iface gbe0 inet manual

auto gbe1
allow-hotplug gbe1
iface gbe1 inet static
        address 192.168.2.1/24
        gateway 192.168.2.2
```

5. Remount the file system as read-only:

```
root@smart-sfp:/home/user # mount -o ro,remount /
```

6. To apply the settings, reboot the module using the "/sbin/reboot" command.

Attention! If you change the network interface settings, management of the module may be lost.

7.1.2. Line interface is connected

Access to the module is possible through ports A and B in this connection and configuration option. No user traffic is transmitted through the probe.

To configure this operating mode, you should assign an IP address on the gbe0 interface (port A) and the gbe1 interface (port B), and delete or comment out the lines with the br0 interface settings:

1. Connect to the device via SSH using the IP address of the br0 interface.

2. Log in as user and then switch to the root account:

```
user@smart-sfp:~$ su
```

3. Remount the file system for writing:

```
root@smart-sfp:/home/user# mount -o remount,rw /
```

4. Edit the /etc/network/interfaces.d/gbe file as follows:

```
root@smart-sfp:/home/user # vi /etc/network/interfaces.d/gbe
auto gbe0
allow-hotplug gbe0
iface gbe0 inet static
        address 192.168.1.1/24
        gateway 192.168.1.2

auto gbe1
allow-hotplug gbe1
iface gbe1 inet static
        address 192.168.2.1/24
```

5. Remount the file system as read-only:

```
root@smart-sfp:/home/user # mount -o ro,remount /
```

6. To apply the settings, reboot the module using the "/sbin/reboot" command.

Attention! If you change the network interface settings, management of the module may be lost.

7.2. In-line connection

7.2.1. Using bridge for data transfer

Access to the module is possible via both ports (A and B) in this connection and configuration option. All user traffic will be transmitted in bridge mode.

By default, interfaces gbe0 (port A) and gbe1 (port B) are already combined into the bridge br0. To configure this operation mode, you should assign an IP address on the br0 interface:

1. Connect to the device via SSH using the IP address of the br0 interface.

2. Log in as user and then switch to the root account:

```
user@smart-sfp:~$ su
```

3. Remount the file system for writing:

```
root@smart-sfp:/home/user# mount -o remount,rw /
```

4. Edit the /etc/network/interfaces.d/gbe file as follows:

```
root@smart-sfp:/home/user # vi /etc/network/interfaces.d/gbe
auto gbe0
allow-hotplug gbe0
iface gbe0 inet manual

auto gbe1
allow-hotplug gbe1
iface gbe1 inet manual

auto br0
allow-hotplug br0
iface br0 inet static
    address 192.168.2.123/24
    gateway 192.168.2.254
    bridge_ports gbe0 gbe1
```

5. Remount the file system as read-only:

```
root@smart-sfp:/home/user # mount -o ro,remount /
```

6. To apply the settings, reboot the module using the “/sbin/reboot” command.

Attention! If you change the network interface settings, management of the module may be lost.

7.2.2. Using routing for data transfer

Access to the module is possible through ports A and B in this connection and configuration option. User traffic is transmitted in routing mode.

To configure this operating mode, you should assign an IP address on the gbe0 interface (port A) and the gbe1 interface (port B), delete or comment out the lines with the br0 interface settings, and also enable routing:

1. Connect to the device via SSH using the IP address of the br0 interface.

2. Log in as user and then switch to the root account:

```
user@smart-sfp:~$ su
```

3. Remount the file system for writing:

```
root@smart-sfp:/home/user# mount -o remount,rw /
```

4. Edit the /etc/network/interfaces.d/gbe file as follows:

```
root@smart-sfp:/home/user # vi /etc/network/interfaces.d/gbe
auto gbe0
allow-hotplug gbe0
iface gbe0 inet static
        address 192.168.1.1/24
        gateway 192.168.1.2

auto gbe1
allow-hotplug gbe1
iface gbe1 inet static
        address 192.168.2.1/24
```

5. Edit the /etc/sysctl.conf file. Add this string to it:

```
net.ipv4.ip_forward=1
```

6. Remount the file system as read-only:

```
root@smart-sfp:/home/user # mount -o ro,remount /
```

7. To apply the settings, reboot the module using the “/sbin/reboot” command.

Attention! If you change the network interface settings, management of the module may be lost.

8. Examples of typical settings

8.1. Disabling RX_LOS signal generation

If there is no signal at the input of the module optical interface (port A), the module generates the RX_LOS signal. When receiving the RX_LOS signal, some host devices will disable their interface. This may prevent the use of the module without a connected line interface, so in this case, it is necessary to disable the generation of the RX_LOS signal. To do this, you need to:

1. Connect to the module via SSH as user and then switch to the root account:

```
user@smart-sfp:~$ su
```

2. Remount the file system for writing:

```
mount -o remount,rw /
```

3. Open the /etc/rc.local file in the “vi” text editor:

```
vi /etc/rc.local
```

4. Add lines:

```
echo "0" > /sys/devices/platform/host-gpio/rx-los-state/value
echo "1" > /sys/devices/platform/host-gpio/rx-los-en/value
```

The lines need to be added before the “exit 0” line.

5. Save changes and close the file:

```
:wq
```

6. Remount the file system as read-only:

```
mount -o remount,ro /
```

7. To apply the settings, reboot the module:

```
/sbin/reboot
```

8.2. Setting up DNS server address

To configure the IP address of the DNS server, you need to:

1. Connect to the module via SSH as user and then switch to the root account:

```
user@smart-sfp:~$ su
```

2. Remount the file system for writing:

```
mount -o remount,rw /
```

3. Open the /etc/rc.local file in the “vi” text editor:

```
vi /etc/resolv.conf
```

4. Add lines with the required DNS server IP addresses, for example:

```
nameserver 8.8.8.8
nameserver 8.8.4.4
```

5. Save changes and close the file:

```
:wq
```

6. Remount the file system as read-only:

```
mount -o remount,ro /
```

8.3. Installing Docker

The module can run applications in Docker containers. To do this, you need to install Docker.
To install Docker, the module should be able to access the internet:

- configure the network settings according to the section 7;
- configure the IP address of the DNS server as described in the section 8.2.

After that, to install Docker you need to:

1. Connect to the module via SSH as user and then switch to the root account:

```
user@smart-sfp:~$ su
```

2. Remount the file system for writing:

```
root@smart-sfp:~# mount -o remount,rw /
```

3. Set iptables as the default firewall:

```
root@smart-sfp:~# update-alternatives --set iptables /usr/sbin/iptables-legacy
```

4. Update the package list:

```
root@smart-sfp:~# apt-get -qq update >/dev/null
```

5. Install the ca-certificates and curl packages:

```
root@smart-sfp:~# DEBIAN_FRONTEND=noninteractive \
apt-get -y -qq install ca-certificates curl >/dev/null
```

6. Create the directory /etc/apt/keyrings:

```
root@smart-sfp:~# install -m 0755 -d /etc/apt/keyrings
```

7. Download Docker GPG key:

```
root@smart-sfp:~# curl -fsSL "https://download.docker.com/linux/debian/gpg" -o \
/etc/apt/keyrings/docker.asc
```

8. Make the GPG key readable by all users:

```
root@smart-sfp:~# chmod a+r /etc/apt/keyrings/docker.asc
```

9. Add the Docker repository to the apt sources list:

```
root@smart-sfp:~# echo "deb [arch=arm64 signed-by=/etc/apt/keyrings/docker.asc] \
https://download.docker.com/linux/debian bookworm test" > \
/etc/apt/sources.list.d/docker.list
```

10. Update the package list again:

```
root@smart-sfp:~# apt-get -qq update >/dev/null
```

11. Clear apt cache:

```
root@smart-sfp:~# apt-get clean
```

12. Install Docker Engine, CLI, and container runtime:

```
root@smart-sfp:~# DEBIAN_FRONTEND=noninteractive apt-get --no-install-recommends \
-y -qq install docker-ce docker-ce-cli containerd.io >/dev/null
```

13. Clear apt cache:

```
root@smart-sfp:~# apt-get clean
```

14. Install Docker Compose:

```
root@smart-sfp:~# DEBIAN_FRONTEND=noninteractive apt-get -y \
-qq install docker-compose-plugin >/dev/null
```

15. To test the installation, run the test container:

```
root@smart-sfp:~# docker run hello-world
```

9. Factory reset

Note. After performing this procedure, all settings and changes in the file system (for example, changes in configuration files) and all installed packages will be lost.

To reset the module to the factory settings:

1. Eject the module from the host equipment (see Section 6.1).
2. Set the DIP switch to the “Recovery mode” position (see Table 3.3).
3. Insert the module into the host device. The factory reset procedure will start automatically. It takes about 15 minutes.
4. Eject the module and set the DIP switch to the “Regular mode” position (see Table 3.3).

10. Software upgrade

10.1. Upgrading from software versions 1.4.5 and later to newer ones

10.1.1. Necessary files

To perform a remote software upgrade, you need to contact technical support to obtain an archive with the necessary files:

- update.tar.gz
- u-boot-env.bin
- updater.sh
- complete_update.sh
- config_file

If the bootloader needs to be upgraded during the upgrade process, the archive will also contain the flash-image.bin file.

10.1.2. Keeping settings during software upgrade

It is possible to perform software upgrade and keep necessary settings and files. Files and directories that should be kept on Smart SFP without removing or setting them to default must be listed in file config_file. By default, config_file has following list of files and directories in it:

```
/etc/passwd
/etc/shadow
/etc/group
/etc/network/interfaces
/etc/network/interfaces.d/
/etc/ssh/
/var/lib/ntpd/ntp.drift
/etc/cron.d/
/etc/cron.daily/
/etc/cron.hourly/
/etc/cron.monthly/
/etc/cron.weekly/
/etc/crontab
```

By using default config_file the following settings will be kept after the software upgrade procedure: users (including passwords, home directory, access rights, interpreter after login), network settings, SSH keys and settings, crontab schedule. If you don't want some of these settings to be kept, you have to remove corresponding lines from config_file. For example, if you don't want to keep users, you have to remove lines:

```
/etc/passwd
/etc/shadow
/etc/group
```

In this case default users will be created after software upgrade.

If you want to keep any other files or directories, you should add them to config_file:

```
/etc/passwd
/etc/shadow
/etc/group
/etc/network/interfaces
/etc/network/interfaces.d/
/etc/ssh/
/var/lib/ntpd/ntp.drift
/etc/cron.d/
/etc/cron.daily/
/etc/cron.hourly/
/etc/cron.monthly/
/etc/cron.weekly/
/etc/crontab
/user_directory/
/var/user_directory/
/user_file
```

For correct software upgrade you must not add directories / and /var in config_file. If you want to keep user files in these directories, you should add paths to exact objects you want to keep, for example: /user_directory/, /var/user_directory/, /user_file.

10.1.3. Upgrade procedure

1. Extract files from the archive received from technical support.
2. Modify the config_file according to chapter 10.1.2 if it is needed.
3. Upload the resulting set of files to Smart SFP into /home/user directory using SFTP or SCP protocols. You should use user account for this.
4. Connect to the module as user via SSH.
5. Switch to root account using the command su.
6. Change the present working directory to /home/user, if it is not it already:

```
root@smart-sfp:~# cd /home/user
root@smart-sfp:/home/user#
```

7. Run commands:

```
root@smart-sfp:/home/user# chmod +x updater.sh
root@smart-sfp:/home/user# chmod +x complete_update.sh
```

8. Start the upgrade procedure:

```
root@smart-sfp:/home/user# ./updater.sh -a ./update.tar.gz -c ./config_file -e ./u-boot-env.bin -d 3
```

To perform software upgrade and reset all settings and files to the factory default state, you should start the software upgrade procedure without config_file in the command:

```
root@smart-sfp:/home/user# ./updater.sh -a ./update.tar.gz -e ./u-boot-env.bin -d 3
```

In this case you have to confirm software upgrade with factory reset by pressing Y, then Enter.

```
You do not provide config file (-c). Are you sure?
Print yes/y or no/n
y
```

All settings, including network settings will be reset. Use default IP addresses to connect to Smart SFP after software upgrade.

9. The software upgrade procedure will take around 10 minutes. The message will appear after it finishes:

```
Please reboot system and then run complete_update.sh script
```

10. If it is necessary to upgrade the bootloader, run these commands:

```
root@smart-sfp:/home/user# echo 0 > /sys/block/mmcblk0boot0/force_ro
root@smart-sfp:/home/user#  dd if=./flash-image.bin of=/dev/mmcblk0boot0 bs=512
```

11. After that reboot Smart SFP:

```
root@smart-sfp:/home/user# reboot
```

Important notice: Next steps must be done fast. After reboot the 86 second watchdog timer will be started. If you will not run complete_update.sh script within this time, rollback procedure will start: Smart SFP will reboot again, and will return to the state before upgrade. This rollback function provides reliability in cases of upgrade failure, lost remote access due to accidental factory reset, or configuration incompatibility with new software version.

12. Connect again to the module as user via SSH

13. Switch to root account using the command su.

14. Change the present working directory to /home/user, if it is not it already:

```
root@smart-sfp:~# cd /home/user
root@smart-sfp:/home/user#
```

15. Run command to finish the upgrade procedure:

```
root@smart-sfp:/home/user# ./complete_update.sh 3
Update completed
```

16. Check the version of installed software using the command:

```
user@smart-sfp:~$ cat /etc/issue
```

10.2. Upgrading from software version 1.4.2 to 1.4.5

10.2.1. Necessary files

To perform a software upgrade, you need to contact technical support to obtain an archive with the necessary files:

- Image
- fdt.dtb
- flash-image.bin
- initramfs.gz.u-boot
- recovery.bin.zip

10.2.2. Module settings after upgrading from software version 1.4.2 to 1.4.5

After the upgrade is complete, all module settings will be reset. The information required to connect to the module is described in section 6. Network interfaces should be configured in accordance with section 7.

10.2.3. Upgrade procedure

Before updating, make sure that the host device in which the module is installed has a stable power supply. If the update is interrupted due to a power outage, the procedure will need to be repeated.

1. Extract files from the archive received from technical support.

2. The resulting set of files need to be uploaded via SFTP or SCP protocols to the Smart SFP in the /home/user directory. The user account should be used to connect.

3. Connect to the module as user via SSH.

4. Switch to root account using the command su.

5. If the current directory is different from /home/user, go to it using the command:

```
root@smart-sfp:~# cd /home/user
root@smart-sfp:/home/user#
```

6. Prepare the partition for the update and mount it by running the commands:

```
root@smart-sfp:/home/user# PATH="/sbin:$PATH"
root@smart-sfp:/home/user# echo -e "d\n\nn\n6156288\n\nw\n" | fdisk /dev/mmcblk0
root@smart-sfp:/home/user# mkfs.ext4 /dev/mmcblk0p8
root@smart-sfp:/home/user# mount /dev/mmcblk0p8 /mnt/
```

7. Copy the update files to the created partition:

```
root@smart-sfp:/home/user# cp Image fdt.dtb flash-image.bin \
initramfs.gz.u-boot recovery.bin.zip /mnt
```

8. Delete update files from the /home/user directory:

```
root@smart-sfp:/home/user# rm Image fdt.dtb flash-image.bin \
initramfs.gz.u-boot recovery.bin.zip
```

9. Unmount the created partition:

```
root@smart-sfp:/home/user# umount /mnt
```

10. Shut down the module using the command:

```
root@smart-sfp:/home/user# poweroff
```

11. Eject the module from the host device (see Section 6.1).

12. Set the DIP switch to the "Recovery" position (see Table 3.3).

13. Insert the module into the host device. The software upgrade procedure will begin, which will take about 15 minutes. During the upgrade, the LED indicator will flash red.

14. When the LED indicator starts to light up green continuously, eject the module and set the DIP switch to the "Regular" position (see Table 3.3).

15. The upgrade procedure is complete; the module has default settings and can be installed back into the equipment.

11. EEPROM memory reprogramming

The module supports EEPROM reprogramming at address A0h (0x50). It is possible in two ways:

1. Using typical EEPROM programmers for SFP modules
2. Without using an EEPROM programmer

11.1. EEPROM reprogramming using a programmer

By default, the module's EEPROM memory is write-protected. To remove this protection, you need to:

1. Connect to the module as user via SSH and then switch to the root account:

```
user@smart-sfp:~$ su
```

2. Remount the file system for writing:

```
mount -o remount,rw /
```

3. Open the /etc/rc.local file in the “vi” text editor:

```
vi /etc/rc.local
```

4. Add the line:

```
echo 0 > /sys/devices/platform/cpu-gpio/host-eeprom-write-protect/value
```

The line needs to be added before the “exit 0” line.

5. Save changes and close the file:

```
:wq
```

6. Remount the file system as read-only:

```
mount -o remount,ro /
```

7. Reboot the module:

```
/sbin/reboot
```

After turning on the module, writing to the EEPROM will not be available immediately, but as the Linux OS boots (10-20 s).

8. Connect to the module and make sure that the value of “host-eeprom-write-protect” is 0:

```
cat /sys/devices/platform/cpu-gpio/host-eeprom-write-protect/value
```

9. Now you can disconnect the module from the network equipment, connect it to the SFP programmer and start reprogramming it as a typical SFP module.

After programming is completed, it is necessary to restore the write protection of the EEPROM memory:

1. Connect to the module as user via SSH and then switch to the root account:

```
user@smart-sfp:~$ su
```

2. Remount the file system for writing:

```
mount -o rw,remount /
```

3. Open the /etc/rc.local file in the “vi” text editor:

```
vi /etc/rc.local
```

4. Delete the line:

```
echo 0 > /sys/devices/platform/cpu-gpio/host-eeprom-write-protect/value
```

5. Save changes and close the file:

```
:wq
```

6. Remount the file system as read-only:

```
mount -o remount,ro /
```

7. Reboot the module:

```
/sbin/reboot
```

11.2. EEPROM reprogramming without using a programmer

To change EEPROM without a programmer, you need to:

1. Copy the EEPROM image that needs to be installed to the /home/user directory of the module. You can use the scp utility or the WinScp program with a graphical interface for this:

```
scp /Users/user/Downloads/image.img user@192.168.1.1:
```

2. Connect to the module as user via SSH and then switch to the root account:

```
user@smart-sfp:~$ su
```

3. Remount the file system for writing:

```
mount -o rw,remount /
```

4. Unlock the EEPROM change by running the command:

```
echo 0 > /sys/devices/platform/cpu-gpio/host-eeprom-write-protect/value
```

5. Run the command:

```
echo 24c02 0x50 > /sys/bus/i2c/devices/i2c-0/new_device
```

6. Change directory to /home/user/ where the image.img file is located:

```
cd /home/user/
```

7. Create a backup copy of the original EEPROM memory contents:

```
/home/admin# cat /sys/bus/i2c/devices/0-0050/eeprom > backup.img
```

8. Rewrite the contents of the module's EEPROM memory to the contents of the image:

```
/home/admin# cat image.img > /sys/bus/i2c/devices/0-0050/eeprom
```

9. Remount the file system as read-only:

```
mount -o remount,ro /
```

10. Reboot the module:

```
/sbin/reboot
```

11. If after rebooting the module is not detected and the connection on port B is not established, then the module should be removed from the switch and reinstalled after 10 seconds.

12. Maintenance

A fully assembled and well operated device does not need any maintenance.

Appendix 1. Pins of the electrical connector of SFP module

Table 12.1. SFP pin descriptions

Pin	Symbol	Name/Description
1	V_{EET}	Transmitter Ground (Common with Receiver Ground)
2	T_{FAULT}	Transmitter Fault.
3	T_{DIS}	Transmitter Disable. Laser output disabled on high or open.
4	MOD_DEF(2)	Module Definition 2. Data line for Serial ID.
5	MOD_DEF(1)	Module Definition 1. Clock line for Serial ID.
6	MOD_DEF(0)	Module Definition 0. Grounded within the module.
7	Rate Select	No connection required
8	LOS	Loss of Signal indication. Logic 0 indicates normal operation.
9	V_{EER}	Receiver Ground (Common with Transmitter Ground)
10	V_{EER}	Receiver Ground (Common with Transmitter Ground)
11	V_{EER}	Receiver Ground (Common with Transmitter Ground)
12	RD-	Receiver Inverted DATA out. AC Coupled
13	RD+	Receiver Non-inverted DATA out. AC Coupled
14	V_{EER}	Receiver Ground (Common with Transmitter Ground)
15	V_{CCR}	Receiver Power Supply
16	V_{CCT}	Transmitter Power Supply
17	V_{EET}	Transmitter Ground (Common with Receiver Ground)
18	TD+	Transmitter Non-Inverted DATA in. AC Coupled.
19	TD-	Transmitter Inverted DATA in. AC Coupled.
20	V_{EET}	Transmitter Ground (Common with Receiver Ground)